



Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2025-26

Class:	BE	Semester:	VIII
Course Code:		Course Name:	RL

Name of Student:	BARI ANKIT VINOD
Roll No. :	61
Assignment No.:	6
Title of Assignment:	
Date of Submission:	
Date of Correction:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Completeness	5	04
Demonstrated Knowledge	3	02
Legibility	2	02
Total	10	08

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Completeness	5	3-4	1-2
Demonstrated Knowledge	3	2	1
Legibility	2	1	0

Checked by

Name of Faculty : Mrs. Sneha Yadav

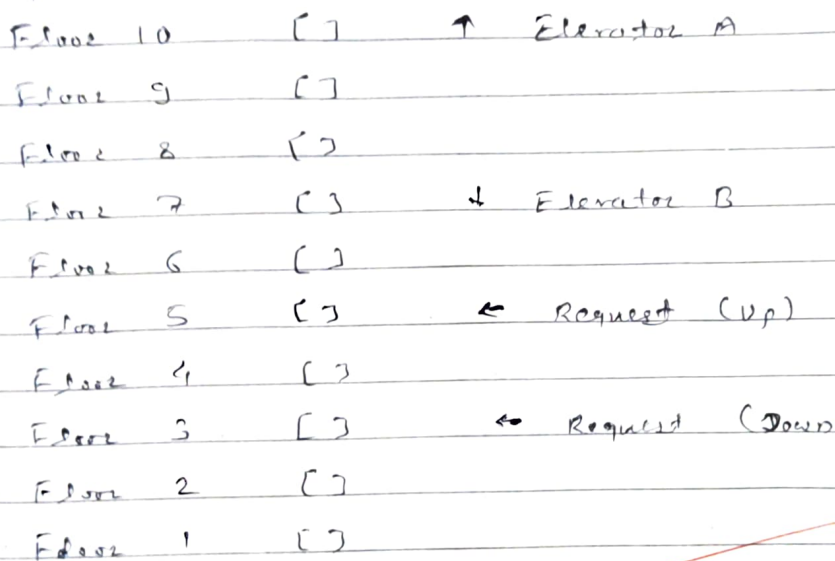
Signature :

Date :

1) demonstrate the concepts of elevator dispatching in a multi-floor building with a suitable diagram, discuss the objectives of the elevator dispatching system and analyze the major challenges involved in designed such a system.

- Elevator dispatching is a process of
- assigning elevators to floor requests
 - deciding movements
 - optimizing passenger wait and travel time.

Simple diagram -



- Request at floor 5 (Up) & request at floor 3 (Down)
- system decides - elevator A - handle upward request
- elevator B - handle downward request

- Objectives of elevator dispatching -
- minimize waiting time
- minimize travel time
- maximize throughput
- energy efficiency

Q. 2) Identify how a scheduling algorithm can be applied to optimize delivery routes, minimize delivery times, and allocate resources effectively to meet customer demands while reducing operational costs.

→ A scheduling algorithm helps -

- assign deliveries to vehicles
- decide optimal routes
- allocate drivers / resources

- Real world scenario -

- multiple delivery orders
- limited vehicles
- different delivery locations and time windows.

- How scheduling is applied -

- ① Route optimization - shortest path (Dijkstra)
 - Travelling Salesman problem
 - Vehicle Routing problem.

- ② delivery time minimization - prioritize urgent deliveries
 - earliest deadlines first.

- ③ Resources allocation - drivers, vehicles, fuel usage.

- ④ load balancing - distribute work evenly across vehicles.

- ⑤ dynamic scheduling - Adjust routes in real-time
 - Traffic
 - New orders
 - Delays.